



Modern CAD Design Plan & Work Tracker

Close the gap from FreeCAD toward SolidWorks and Fusion 360, starting on the FreeCAD foundation and shipping first on Apple Silicon.

Repository **brucecodes2023/FreeCAD_BT**
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Branch **cursor/modern-cad-phase1-953c**

Updated

HOW TO USE THIS DOCUMENT

This is the working design record for the fork. IDs are stable. When work lands, flip the status tag, name the files, and add a line to the work log at the end. The original Cloud Agent scoping note (`docs/macos-native-ui-scope.tex` on branch `cursor/setup-cloud-agent-env-d2a0`) covered A/B/C only. This file keeps those IDs and adds shell (D), look (E), modeling UX (F), and simulation (G).

GitHub does not render `.tex`. Keep the Markdown edition open in a browser tab while we work: `Documents/README.md` (also linked from repo-root `PLAN.md`). The PDF in this folder is the typeset copy; CI rebuilds it as the **Design PDF** Action artifact when this source changes.

Constraints that still hold

- Source-only until a Mac checkout builds and launches the GUI. Do not treat a Linux VM as a visual preview.
- Work from FreeCAD's existing extension points (preference packs, YAML/QSS, workbench `ToolBarItem` trees, Elmer equation objects). Do not start with a Metal rewrite or a ribbon-only skin.
- Effort ratings describe technical scope, not calendar time.

Product thesis

SolidWorks is the target *feel* (FeatureManager, CommandManager-like strip, familiar mouse, guided simulation). Fusion is the target *shell* (ribbon while modeling, dashboard on launch). FreeCAD is the *foundation* (OCCT kernel, Body/Tip history, workbenches, CalculiX + Elmer). The gap is closed in this order, because that is the order the codebase actually supports:

1. **Shell** — Home dashboard + Fusion-style ribbon over existing commands.
2. **Look** — YAML tokens, QSS, icons. Not a C++ widget rewrite.

3. **Modeling UX** — Body/Tip, naming, rollback, sketch-in-tree.
4. **Mac smoothness** — Qt/OpenGL mitigation now; Metal later.
5. **Simulation** — guided CalculiX plus Elmer first-principles as WIP.

Keep the native macOS menu bar. The ribbon is the command strip, not a replacement for File / Edit / View.

Status tracker

SHIPPED AND NATIVE MAC

ID	Item	Status	Notes
A1	SolidWorks nav default	Done	New profiles only; existing <code>User.cfg</code> unchanged.
A2	Modern CAD behavior pack	Done	Nav, cube, combo view, ribbon flag.
A3	View cube de- faults	Done	Top-right, size 132.
A4	First-start on- boarding	Done	Applies Modern CAD pack once.
A5	Fusion 360 nav class	Done	Additive; MMB pan / Shift+MMB orbit. Default stays SolidWorks.
A6	Fusion-style ribbon	Partial	QAT, overflow, captions, sketch contextual tabs. Mac QA remaining.
B0	Traffic lights already native	Done	Finding; no code move.
B1	Unified title + toolbar	Done	<code>setUnifiedTitleAndToolBarOnMac(true)</code> .
B2	Native dock/- toolbar chrome	Done	Cocoa chrome; <code>macos-native-chrome.qss</code> .
B3	NSWindow full- size content	Later	New <code>.mm</code> shim.
B4	Bundle id + dark mode	Done	<code>org.freecad.FreeCAD</code> ; Aqua appearance off.
C1	arm64 + ma- cOS 11 pin	Done	CMake helpers + conda-macos preset.
C2	Optional LTO / apple-m1	Done	Off by default.
C3	Enable TBB	Done	<code>HAVE_TBB</code> + <code>ImportOCAF</code> .
C4	Defer Init- Gui.py	Later	Startup ordering risk.
C5	Metal / Qt RHI viewport	Later	Real M5 fix. Near-term: vsync + software GL.

SHELL, LOOK, MODELING, SIMULATION

ID	Item	Status	Notes
D1	Home dashboard	Partial	Metrics, tips, command search, projects, new-file cards.
D2	Project folders	Partial	Create + list; open file from folder.
D3	Dashboard health widgets	Partial	Drawing + Continue; recompute errors; GPU warning; search.
E1	Modern CAD Light/-Dark tokens	Partial	Cool gray + SW blue; iterate on Mac.
E2	Denser Feature tree / icons	Partial	Pack: 32px toolbars, tree icon 24 / indent 10.
E3	Viewport / PD pre-view colors	Partial	Gradient off; dress-up preview blue-gray.
E4	Custom sketch/feature icons	Later	Optional icon set.
F1	Rollback bar	Later	Body/Tip history UX.
F2	Topological naming UX	Later	Kernel work, not chrome.
F3	Sketch-in-tree FeatureManager	Later	Single tree like SW.
F4	In-context assembly edits	Later	Beyond Ondsels joints.
G1	First Principles study stub	WIP	Elmer elasticity + heat; labeled WIP.
G2	Guided equation wizard	Partial	Mesh / material / BC checklist before solve.
G3	CalculiX study pre-sets	Partial	Static + thermal wrappers; no auto-mesh.
G4	Assembly contact FEA	Later	Large gap vs SW Simulation.

Workstream A CAD-familiar defaults

Goal: a first launch that feels like SolidWorks / Fusion without touching preferences. Vehicle: built-in preference pack plus shipped fallbacks.

A1 — SolidWorks navigation

Navigation styles live under `src/Gui/Navigation/`. `SolidWorksNavigationStyle` already existed. New profiles now fall back to it, with turntable orbit and object-center rotation (`DefaultNavigationStyleName`, `DefaultOrbitStyle`, `DefaultRotationMode` in `NavigationStyle.h`). Existing `User.cfg` is not migrated.

A2 — Modern CAD behavior pack

Pack at `src/Gui/PreferencePacks/Modern CAD/`: SolidWorks nav, cube, combo-view layout, `UseRibbon=1`. Type is Behavior, not Theme — apply on top of Light/Dark.

A3 — View cube

NaviCube top-right, size 132, rotate-to-nearest. Configurable via `BaseApp/Preferences/Nav-iCube`.

A4 — First-start onboarding

Start workbench `FirstStart2024` flow applies the Modern CAD pack once (`ModernCADDefaultsApplied`), skipped in GUI tests. Copy mentions SW/Fusion nav and the ribbon/dashboard.

A5 — Fusion 360 navigation class

`Gui::Fusion360NavigationStyle` is registered. Fusion defaults: MMB pan, Shift+MMB orbit, scroll zoom. New profiles still use SolidWorks. Pick Fusion 360 in Preferences → Display → Navigation.

A6 — Fusion-style ribbon

Originally deferred. Product decision reversed: ribbon *replaces* classic toolbars while a document is open.

Shipped

- `Gui::RibbonBar / RibbonManager` consume the workbench `ToolBarItem` tree.
- Tabs = toolbars; groups split on separators; workbench combo on the tab row.
- Hidden when the active MDI view is `StartView`.
- Classic `QToolBars` hidden while ribbon is on (including after sketch `setState`).
- Preference `BaseApp/Preferences/MainWindow/UseRibbon` (default on; off in GUI tests).
- Ribbon sits above the MDI area, not inside the 22 px unified title bar.
- Quick Access strip: New, Save, Undo, Redo.
- Per-page overflow chevron (») when groups do not fit.
- Fusion-sized captions (toolbar icon size from prefs, text under icon).
- Contextual tabs: `Unavailable` (sketch edit) toolbars shown only while a sketch is in edit.

Still to polish on Mac: height vs traffic lights, overflow at laptop widths, optional QAT customization.

Workstream D Home dashboard

The landing page must *not* show the ribbon. It is a dashboard.

D1 — Dashboard chrome

`StartGui::StartView` documents page is now Home: title, three metric cards (recent files, project count, graphics backend), then projects, new-file cards, recent files, examples. Metrics are placeholders for later health/solver widgets.

D2 — Projects

`Start::ProjectsModel` stores folder paths under `BaseApp/Preferences/Mod/Start/Projects`. New Project picks a parent folder, creates a named directory and a `.freecad-project` JSON marker, then lists the project. Clicking opens a file dialog in that folder. Not a SolidWorks assembly project yet — a workspace folder.

D3 — Dashboard widgets (partial)

- Templates: Part Design Body, Assembly, Drawing (TechDraw), plus Continue last file.
- Document health: recompute error count across open documents.
- In-page command search (`Gui::CommandCompleter`).
- Units / navigation style on a metric card.
- What's-new / first-run tips (dismissible).
- Mac GPU warning when the OpenGL renderer string looks like Metal 90.x.

Pinned folders and sketch-solver health remain later.

Workstream E SolidWorks-like facelift

Theming is YAML tokens + `FreeCAD.qss` (`StyleParameters::ParameterManager`), not a C++ skin.

E1 — Modern CAD Light / Dark

New packs and YAML:

- Light: chrome #F5F5F7, accent #0066B3, flat padding, light-gray 3D background.
- Dark: graphite #2D2D30, brighter blue accent.

Apply from Preferences → Preference Packs, on top of the Modern CAD behavior pack. Visual QA waits for a Mac launch.

E2–E4 — Density, viewport, icons

Packs set toolbar icons to 32 px and denser Feature tree (icon 24, indent 10, item spacing 1). Viewport gradient stays off. PartDesign dress-up preview is blue-gray instead of magenta (`StyleParameters.h`). Custom SW-like glyphs (E4) remain later.

Workstream F Modeling kernel UX

This is the real SolidWorks gap. Chrome cannot fake it.

Today. PartDesign is Body + Tip + history. Pad/Pocket/Fillet go through ProfileBased / DressUp. Recompute is `App::Document::recompute()` topo-sort. Assembly is `AssemblyObject` + `OndselSolver` joints (kinematic), not SW mates.

Gaps vs SolidWorks

- F1 — no real rollback bar on the history tip.
- F2 — topological naming is still fragile after face edits.
- F3 — sketch is a separate object; no single FeatureManager tree.
- F4 — no in-context assembly edits / SW-style mates driving geometry.

Do not schedule F before D/E are usable on a Mac. When F starts, work in `src/Mod/PartDesign/` and `src/App/Document.cpp`, not in QSS.

Workstream G Simulation / first principles

Today. `FemAnalysis` plus external solvers. CalculiX is the structural workhorse. Elmer is the continuum-PDE path (elasticity, heat, flow, EM equation objects under `src/Mod/Fem/`).

G1 — First Principles Study

Command `FEM_FirstPrinciplesStudy` (Model + Solve toolbars/menus): creates or reuses an analysis, adds Elmer, attaches elasticity + heat equations, shows a one-time WIP dialog. Does *not* replace CalculiX, add contact, or invent a SW Simulation study tree.

G2–G4 — Wizard, CalculiX presets, contact

`FEM_StudyGuidedWizard` is a mesh / material / BC checklist (no new solver).

`FEM_CalculiXStaticStudy` and `FEM_CalculiXThermalStudy` create analysis + CalculiX + empty material. Assembly-wide contact (G4) remains a large gap, not a stub.

Workstream B macOS-native chrome

B0 — Traffic lights already native

`Gui::MainWindow` is a plain `QMainWindow`. Qt/Cocoa draws the standard frame; red/amber/green already sit top-left. No code move.

B1 — Unified title + toolbar

`setUnifiedTitleAndToolBarOnMac(true)` in the `MainWindow` constructor. The ribbon is *not* placed in that unified bar (too tall); it is a widget above the MDI area.

B2 — Native dock & floating toolbars

Floating toolbars keep native Cocoa chrome. Overlay `src/Gui/Stylesheets/macos-native-chrome.qss` is appended on `Q_OS_MAC`.

B3 — NSWindow full-size content

Content under the title bar / large-title behavior needs a new Objective-C++ `.mm` shim. Precedent exists (QuickLook, 3Dconnexion) but no window-chrome Cocoa code yet.

B4 — Bundle identity & appearance

`CFBundleIdentifier=org.freecad.FreeCAD`, `NSRequiresAquaSystemAppearance=false`. Signing/notarization is a release concern, not this branch.

Workstream C Apple Silicon & M5 smoothness

Why M4/M5 feel laggy

The 3D view is OpenGL via Coin3D + `QOpenGLWidget` (`src/Gui/Quarter/QuarterWidget.cpp`, `src/Gui/View3DInventorViewer.cpp`). On macOS that OpenGL is Apple's Metal translation layer (OpenGL 2.1 Metal - 90.x). M5 GPUs are Metal-4 oriented; the translation path is the known hitch/crash source (upstream reports on M5 MacBook Air + macOS Tahoe, slow Sketcher recompute on M4). Safe mode does not fix it. Qt 6.8.3 has a related `QMacAccessibilityElement` use-after-free on Tahoe.

More OpenGL tweaks will not close this gap. C5 (Metal / Qt RHI) is the real fix.

C1 — Architecture pin

`CMAKE_OSX_ARCHITECTURES=arm64` when building on Apple Silicon; default
`CMAKE_OSX_DEPLOYMENT_TARGET=11.0`.

C2 — Optional LTO / `-mcpu=apple-m1`

`FREECAD_USE_LTO` and `FREECAD_APPLE_SILICON_TUNING` are off by default.

C3 — TBB

`find_package(TBB)` before compiler checks so `HAVE_TBB` in `src/config.h.cmake` lights up `ImportOCF` parallel paths.

C4 — Faster startup

All modules' `InitGui.py` still run eagerly (`src/Gui/FreeCADGuiInit.py`). Deferral has ordering risk.

C5 — Metal / Qt RHI

Near-term (shipped): `QSurfaceFormat::setSwapInterval(1)` on macOS; software OpenGL fallback tooltip updated for M-series. Prefer Qt newer than 6.8.3 on Tahoe when building. Do not treat nav cube, extra GL contexts, or QSS as the root cause.

Sequencing

Phase	Focus	Items
Phase 1	Familiar defaults	A1, A3, A2, A4 Done
Phase 2	Native Mac chrome + arm64	B1, B2, B4, C1, C2, C3 Done
Phase 3	Shell + look v0	A6, D1, D2, E1, G1 Partial / WIP
Phase 4	Shell/look polish (source in; Mac QA)	A6 polish, D3, E2–E3 Partial
Phase 5	Modeling UX	F1–F4
Phase 6	Simulation depth + Metal	G4, C5, B3, C4 (A5/G2/G3 started)

Phase 3 is the current working phase. Modeling kernel (F) and Metal (C5) wait until a Mac build has validated Phase 3 visually.

Validation plan

Change	How it is proven
A1–A4 defaults / pack	Fresh profile: SW orbit, cube top-right, Modern CAD applied once.
A6 ribbon	Document open → ribbon + QAT; Home → no ribbon; overflow »; sketch edit shows contextual tabs.
D1–D3 dashboard	Home shows metrics, tips, search, New Project, Drawing, Continue.
E1–E3 look	Modern CAD Light/Dark; 32px icons; flat viewport; PD dress-up blue-gray.
G1–G3 FEM	First Principles Elmer stub; Guided Wizard; CalculiX static/thermal presets.
A5 Fusion nav	Preferences lists Fusion 360; default remains SolidWorks.
B1/B2/B4 Mac chrome	Traffic lights top-left; unified toolbar; dark mode follows system.
C1/C3 build	Reports arm64; TBB found; ImportOCAF can use it.
C5 near-term	Vsync on; software GL option documented; M5 still expected to need Metal.
No regressions	FreeCAD CLI + C++ self-tests remain green. GUI tests force ribbon off.

Grounded file index

Concern	Key path(s)
This plan	Documents/modern-cad-design.tex
Navigation defaults	src/Gui/Navigation/NavigationStyle.h, View3DSettings.cpp
Modern CAD packs	src/Gui/PreferencePacks/Modern CAD/ and Light/Dark
Ribbon	src/Gui/RibbonBar.cpp, RibbonManager.cpp, Workbench.cpp
Ribbon QSS	src/Gui/Stylesheets/fusion-ribbon.qss
Home dashboard	src/Mod/Start/Gui/StartView.cpp, ProjectsModel.cpp
Theme tokens	src/Gui/Stylesheets/parameters/Modern CAD Light.yaml
Main window chrome	src/Gui/MainWindow.cpp, ToolBarManager.cpp
macOS bundle	src/MacAppBundle/FreeCAD.app/Contents/Info.plist
arm64 / TBB / LTO	cMake/FreeCAD_Helpers/, SetupTBB.cmake
OpenGL / vsync	src/Gui/Application.cpp, Quarter/QuarterWidget.cpp
FEM first principles	src/Mod/Fem/femcommands/commands.py, femtools/study_presets.py
FEM wizard	src/Mod/Fem/femguiutils/study_wizard.py
Fusion 360 nav	src/Gui/Navigation/Fusion360NavigationStyle.cpp
PartDesign kernel	src/Mod/PartDesign/App/Body.h, src/App/Document.cpp
Assembly	src/Mod/Assembly/App/AssemblyObject.*
GUI startup	src/Gui/FreeCADGuiInit.py, StartupProcess.cpp

Compile and launch on a Mac

Build from source on Apple Silicon with pixi. Do not treat a Linux VM as a visual preview. Full copy-paste lives in Documents/README.md; this is the short form.

Prerequisites. Xcode Command Line Tools (`xcode-select -install`) and pixi (`curl -fsSL https://pixi.sh/install.sh | bash`). Homebrew Qt is not used. Plan on about 20 GB free (clone + pixi env + build/release objects). `install-release` is optional for a first launch. macOS 27 beta is newer than the Qt 6.8.3 pin — save the log if configure or the GUI fails.

```
git clone https://github.com/brucecodes2023/FreeCAD_BT.git
cd FreeCAD_BT
```

```
git checkout cursor/modern-cad-phase1-953c
pixi install
pixi run initialize
pixi run configure-release
pixi run build-release

export FREECAD_USER_HOME="$HOME/freecad-modern-cad-fresh"
mkdir -p "$FREECAD_USER_HOME"
pixi run freecad-release
```

Use a fresh FREECAD_USER_HOME so A1 defaults apply. After launch: Home has no ribbon; opening a document shows the ribbon and QAT; apply Modern CAD Light/Dark on top of the Modern CAD behavior pack; Fusion 360 nav is optional in Preferences. If the 3D view lags, enable software OpenGL and restart. C5 Metal stays later.

Work log

Update this list every time the plan or the code moves. Newest first.

- **13 Aug 2026 — Remaining IDs (source).** A5 Fusion nav; A6 QAT / overflow / contextual tabs; D3 dashboard widgets; E2/E3 density + PD previews; G2 wizard; G3 CalculiX presets. Mac compile walkthrough added. F1–F4, C5 Metal, B3, C4, E4, G4 still later.
- **13 Aug 2026 — Documents/ revived.** Living LaTeX plan replaces the missing docs/macos-native-ui-scope.tex. Status tracker covers A–G.
- **13 Aug 2026 — Phase 3 v0.** Fusion ribbon (A6), Home dashboard + projects (D1/D2), Modern CAD Light/Dark (E1), FEM First Principles Study WIP (G1), macOS vsync + software-GL copy (C5 near-term). PR #2.
- **13 Aug 2026 — Phase 2.** B1 unified toolbar, B2 native chrome, B4 bundle id, C1 arm64 pin, C2 optional LTO/mcpu, C3 TBB.
- **13 Aug 2026 — Phase 1.** A1 SolidWorks nav defaults, A2/A3 pack + cube, A4 first-start apply.
- **Earlier — env-setup branch.** Original A/B/C scoping LaTeX and Cloud Agent environment notes. Not on this branch.

Effort ratings describe technical scope (breadth of components and invasiveness), not calendar estimates. “Exists today” items are configuration/defaults; “new work” items add code behind existing extension points. Compile with `cd Documents && ./build.sh`.